



Random Lipschitz functions on graphs with weak expansion

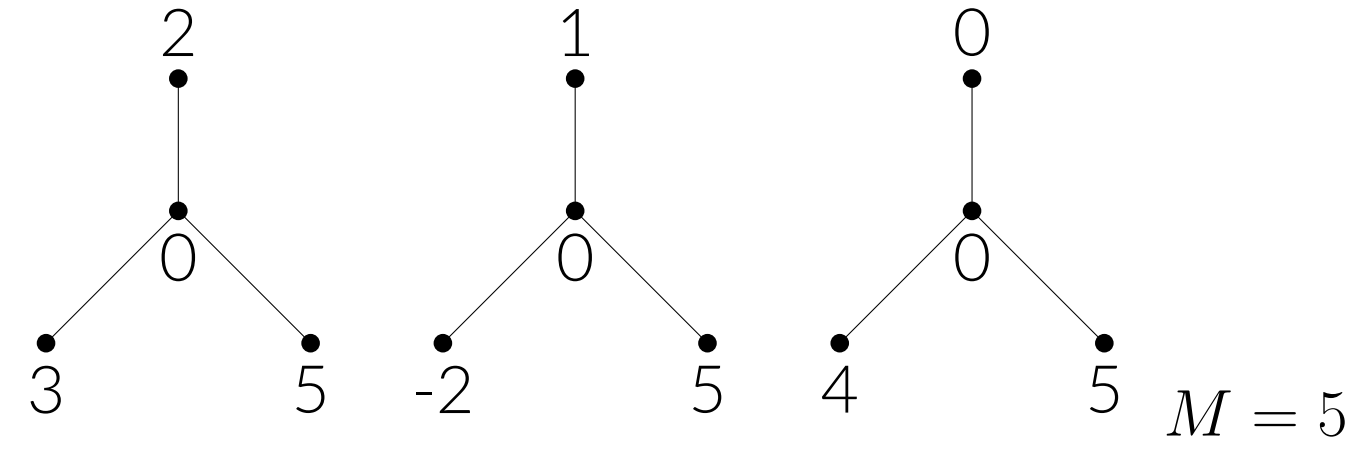
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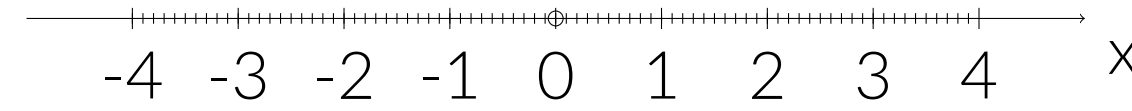


Lipschitz Functions

- Take a graph G with n nodes.
- Let v_0 be a fixed vertex in a graph G and let M be a positive integer.
- Say $\text{Lip}_{v_0}(G; M)$ is the family of maps $f : V(G) \rightarrow \mathbb{Z}$ such that $f(v_0) = 0$ and $|f(u) - f(w)| \leq M$ if $u \sim w$ in G .



- Sample f from $\text{Lip}_{v_0}(G; M)$ **uniformly at random**.
- We want to understand **real-valued Lipschitz functions**!
- Define $\text{Lip}_{v_0}(G; \infty)$ as the family of maps $f : V(G) \rightarrow \mathbb{R}$ such that $f(v_0) = 0$ and $|f(u) - f(w)| \leq 1$ if $u \sim w$ in G .
- Roadmap:** $\lim_{M \rightarrow \infty} \left\{ \frac{f}{M} \mid f \in \text{Lip}_{v_0}(G; M) \right\} = \text{Lip}_{v_0}(G; \infty)$



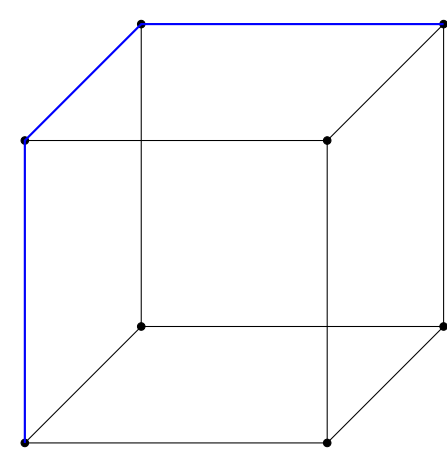
Research Question

The **range** of a (Lipschitz) function f on a connected graph G is

$$R(f) = \sup_{u, v \in G} \{f(u) - f(v)\}$$

Can we give better lower or upper bounds for a *typical* $R(f)$?

- Diameter of a graph gives a natural upper bound.
- We can give tighter bounds that hold with high probability for a uniformly chosen random M -Lipschitz function!



Background

- For a d -dimensional hypercube with n nodes, the diameter is $d = \log_2(n)$ but

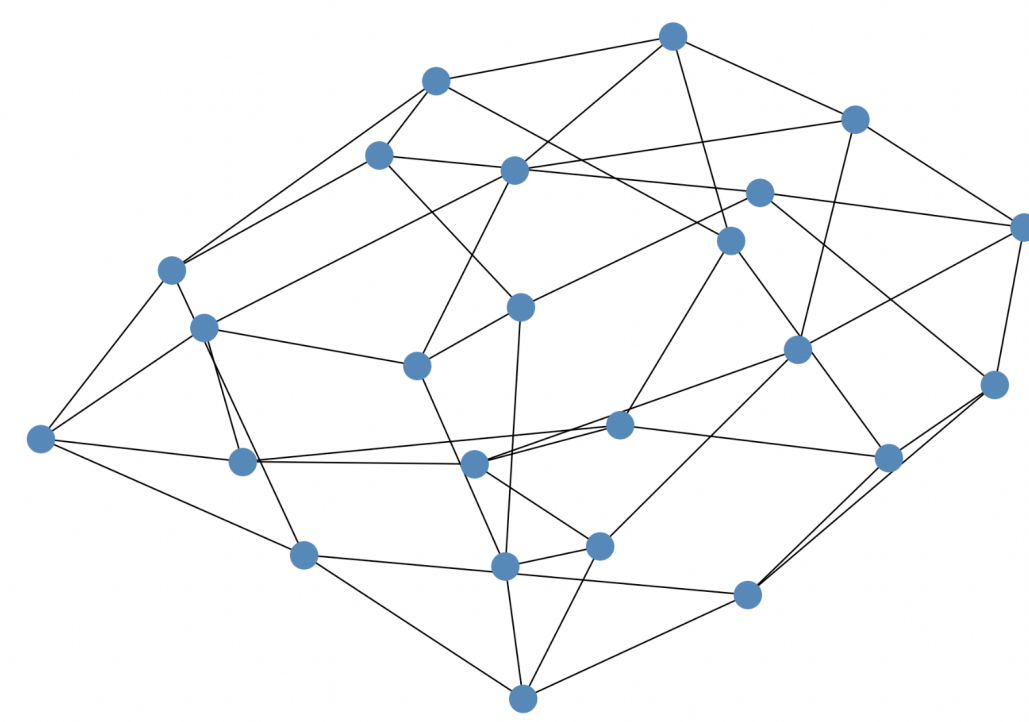
$$\mathbb{P}[R(f) \leq 5] = 1 - o(1) \text{ as } d \rightarrow \infty$$

where f is a random $\mathbb{Z}$ -homomorphism (a type of 1-Lipschitz function) [3] [2].

- For a d -regular λ -good expander, the diameter is $O(\log(n))$

$$\mathbb{P}[R(f) \leq C_{d,\lambda} M \log(\log(n))] = 1 - o(1) \text{ as } n \rightarrow \infty$$

where f is a random M -Lipschitz function [4].^a



^aBecause they considered $\lambda \leq \frac{d}{32(M+1)\log(9Md^2)}$, their results do not generalize to real-valued functions as $M \rightarrow \infty$.

Expansion Criteria

Previously, expanders (graphs with good expansion) were studied: If G is a d -regular λ -expander, then, for *all* r ,

$$|B_{\log(\log(n))}(v)| \geq \log(n)^{2\log\left(\frac{d}{2\lambda}\right)}.$$

Our assumption has the opposite flavor. We assume *low expansion*: For *some* r ,

$$S(G, r) = \sup_{v \in G} |B_r(v)| \leq \log_6(n).$$

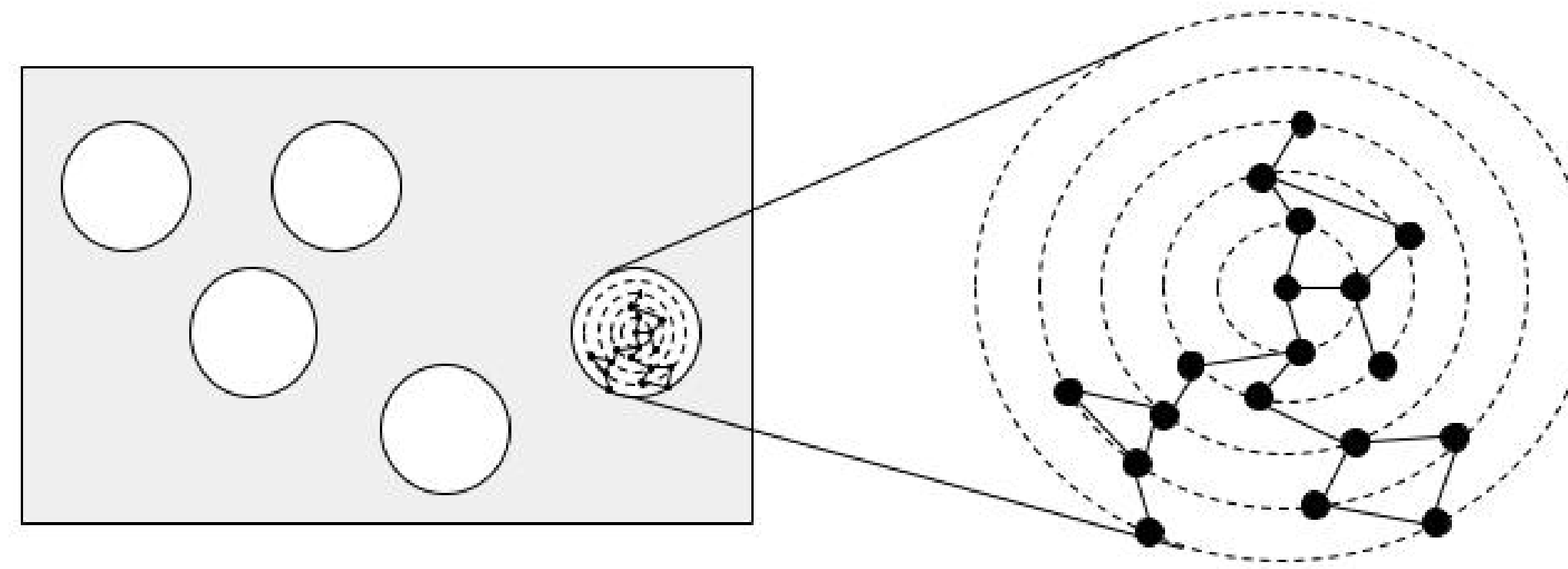
Theorem (I., Park) lower bound

There is a constant $c > 0$ for which the following holds. Let G be a connected graph on n vertices. Fix $v_0 \in V(G)$, and let $f \in_R \text{Lip}_{v_0}(G; M)$. Then for any r that satisfies $\max_{v \in G} |B_r(v)| \leq c \log(n)$,

$$\Pr \left[R(f) \geq \frac{Mr}{2} \right] \rightarrow 1.$$

Proof sketch

- First, use the expansion assumption to conclude that there are many disjoint balls of radius r (proof is by induction).
- Then, lower bound the probability a typical Lipschitz function grows quickly on a ball.



P is the set of Lipschitz functions on the ball with a high range. Q is the set of all Lipschitz functions. We want to lower bound P/Q . Now, it is a counting problem!

To do so, we find a set C and mappings T from C to P and Q such that

1. C partitions P : for all $c \neq c' \in C$ $T_P(c) \cap T_P(c') = \emptyset$

2.

$$\frac{P}{Q} \geq \inf_{c \in C} \frac{T_P(c)}{T_Q(c)}$$

Corollaries

- Graphs with bounded degree:** Sample f from $\text{Lip}_{v_0}(G; M)$ uniformly at random. Then,

$$\Pr \left[R(f) \geq \frac{M}{2} \log_d \log_4 n \right] = 1 - o(1)$$

- Matching lower bound for minimal expanders:**^a Sample f from $\text{Lip}_{v_0}(G; M)$ uniformly at random.

$$\mathbb{P} \left[R(f) \geq \frac{C}{\log(d/\lambda)} M \log \log n \right] = 1 - o(1)$$

for some absolute constant C . (Our constant does not depend on M , which is why we can generalize our results to real-valued functions.)

- Real-valued Lipschitz functions:** Sample f from $\text{Lip}_{v_0}(G; \infty)$ uniformly at random. For any r that satisfies $S(G, r) \leq \log_4 n$, we have

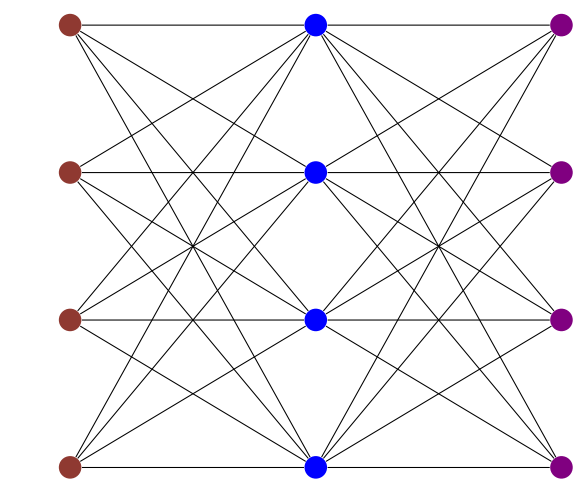
$$\Pr \left[R(f) \geq \frac{r}{2} \right] = 1 - o(1)$$

^aFor any d -regular expander, we know that $S(G, r) \geq \left(\frac{d}{2\lambda}\right)^{2r}$. We call an expander minimal if this inequality is tight.

Local vs. Global Expansion

It is natural to ask how sharp our assumption is. For example, what happens when the degree $d \gg \log(n)$?

- We propose $C_{n,d}$ for this exploration. It is a cycle of n layers, where each layer has d vertices and is connected to both its adjacent layers by a complete bipartite graph.



- Unlike expanders,^a $C_{n,d}$'s initial expansion is large but its growth rate goes to 0 as $r \rightarrow \infty$. Then, one might think that a non-trivial lower bound exists even if $d \gg \log(n)$. On the other hand, we know

$$\Pr[R(f) \leq 3] = 1 - o(1)$$

where f is a random $\mathbb{Z}$ -homomorphism (a type of 1-Lipschitz function) [1].

Theorem (I. Park) upper bound

There is a constant C for which the following holds. For any $\gamma > C$, if $k > \gamma M^2 \log(Mn)$ and, then for $f \in_R \text{Lip}_{v_0}(C_{n,k}; M)$,

$$\Pr(R(f) \leq M) = 1 - O(1/\gamma).$$

Furthermore, if $\log M = o_n(\log n)$, then

$$\Pr(R(f) \leq M) \rightarrow 1.$$

Entropy (as a counting tool)

- The entropy^b $H(X)$ of X is given by

$$H(X) = \sum_x -p(x) \log p(x),$$

- (Maximality of the uniform) For random variable X ,

$$H(X) \leq \log|\text{range}(X)|$$

If X is uniform one can change inequality with equality!

Property (2) makes clear why entropy can be used as an enumeration tool.

Intuition

- Non-constant layers restrict the adjacent layers due to the Lipschitz condition
- Global entropy is maximized when the range of functions on each layer is around M (not too small or large)
- If the range of the function in each layer is $\approx M$, the typical function is flat *globally*.
- Tightness:** If $(M \ll) k \leq \frac{M \ln(Mn)}{2}$, then the *upper bound* doesn't hold. To see this, let α be the number of M -Lipschitz functions in which all vertices take values from $[0, M]$, and β be that with one vertex takes the fixed value $M + 1$ and its neighbors take values from $[1, M]$ (and all the other vertices are in $[0, M]$). Then $\beta/\alpha = ((n - 2)k - 1) \left(\frac{M}{M+1}\right)^{2k} \gg 1$.

References

- [1] Itai Benjamini, Ariel Yadin, and Amir Yehudayoff, *Random graph-homomorphisms and logarithmic degree*, Electron J Probab **12** (2006).
- [2] David Galvin, *On homomorphisms from the hamming cube to $\mathbb{Z}$* , Israel Journal of Mathematics **138** (2003), 189--213.
- [3] Jeff Kahn, *Range of cube-indexed random walk*, Israel Journal of Mathematics **124** (2001), 189--201.
- [4] Ron Peled, Wojciech Samotij, and Amir Yehudayoff, *Lipschitz functions on expanders are typically flat*, Combinatorics, Probability and Computing **22** (2013), no. 4, 566--591.

^aExpanders behave like a tree locally but far from being a tree globally.

^bHelpful to (informally) think of entropy as a measure of the degree of randomness.